

Tarea 3 - Montenegro, Maximiliano

1- Método Storch:

$$T(s) = \frac{1}{\sinh \phi + \cosh \phi}$$

Trabajando en el numerador

$$(\sinh \phi + \cosh \phi) \frac{1}{\sinh \phi} = 1 + \coth \phi$$

$$\frac{\cosh \phi}{\sinh \phi} = \coth \phi = \frac{1}{s} + \frac{\frac{3}{s}}{\frac{5}{s} + \frac{1}{s} + \dots}$$

$n=2$

$$\coth \phi = \frac{1}{s} + \frac{1}{\frac{3}{s}} = \frac{1}{s} + \frac{s}{3} = \frac{s^2 + 3}{3s} = \frac{\cosh \phi}{\sinh \phi}$$

$$T_2(s) = \frac{1(3)}{s^2 + 3s + 3} = \frac{3}{s^2 + 3s + 3} =$$

$$\omega_0^2 = 3 \Rightarrow \omega_0 = \sqrt{3}$$

$$\frac{\omega_0}{Q} = 3 \Rightarrow Q = \frac{\omega_0}{3} = \frac{\sqrt{3}}{3}$$

$$\therefore \frac{\omega_0}{Q} = \frac{\sqrt{3}}{\frac{\sqrt{3}}{3}}$$

$$T_2(s) = \frac{(\sqrt{3})^2}{s^2 + s \frac{\sqrt{3}}{3} + (\sqrt{3})^2}$$

$n=3$

$$\coth \phi = \frac{1}{s} + \frac{1}{\frac{3}{s} + \frac{1}{s}} = \frac{1}{s} + \frac{1}{\frac{3+s}{s}} = \frac{1}{s} + \frac{s}{3+s} = \frac{1}{s} + \frac{1}{\frac{15+s^2}{5s}}$$

$$\coth \phi = \frac{1}{s} + \frac{5s}{s^2 + 15} = \frac{s^2 + 15 + 5s^2}{s(s^2 + 15)} = \frac{6s^2 + 15}{s^3 + 15s} = \frac{\cosh(\phi)}{\sinh(\phi)}$$

$$T_3(s) = \frac{1(15)}{s^3 + 15s + 6s^2 + 15} = \frac{15}{s^3 + 6s^2 + 15s + 15}$$

Mediante un script en python

$$T_3(s) = \frac{15}{s^3 + 6s^2 + 15s + 15} = \frac{2,322}{s + 2,322} \cdot \frac{6,459}{s^2 + s \cdot 3,678 + 6,459}$$

$$\omega_0^2 = 6,459 \Rightarrow \omega_0 = 2,542$$

$$\frac{\omega_0}{Q} = 3,678 \Rightarrow Q = \frac{2,542}{3,678} = 0,691$$

$$T_3(s) = \frac{(12,322)^2}{s + (12,322)^2} \cdot \frac{(2,542)^2}{s^2 + s \cdot \frac{2,542}{0,691} + 2,542^2}$$

$$n=4$$

$$\cot \phi = \frac{1}{s} + \frac{1}{\frac{3}{s} + \frac{1}{\frac{5}{s} + \frac{1}{\frac{7}{s}}}} = \frac{1}{s} + \frac{1}{\frac{3}{s} + \frac{1}{\frac{5}{s} + \frac{1}{\frac{7}{s}}}} = \frac{1}{s} + \frac{1}{\frac{3}{s} + \frac{7s}{s^2 + 35}}$$

$$\cot \phi = \frac{1}{s} + \frac{s^2 + 35s}{3s^2 + 105 + 7s} = \frac{10s^2 + 105 + s^4 + 35s^2}{10s^3 + 105s} = \frac{\cot \phi}{\sin \phi}$$

$$T(s) = \frac{1 \cdot 105}{s^4 + 10s^3 + 45s^2 + 105s + 105} = \frac{105}{s^4 + 10s^3 + 45s^2 + 105s + 105}$$

Mediante script de python

$$T_4(s) = \frac{3,023^2}{s^2 + \frac{3,023}{0,5219}s + 3,023^2} \cdot \frac{3,389^2}{s^2 + \frac{3,389}{0,8055}s + 3,389^2}$$

$$\begin{aligned} \omega_{01} &= 3,023 \\ Q_1 &= 0,5219 \\ \omega_{02} &= 3,389 \\ Q_2 &= 0,8055 \end{aligned}$$

2.

$$n=2$$

$$\alpha_{max} \mid \omega_p = 1$$

$$T_2(s) = \frac{3}{s^2 + 3s + 3}$$

$$|T_2(s=j1)| = \left| \frac{3}{(j1)^2 + 3(j1) + 3} \right| = \frac{3}{|3 - 1 + 3j|} = \frac{3}{|2 + 3j|} = \frac{3}{\sqrt{4 + 9}} = \frac{3}{\sqrt{13}}$$

$$|T_2(j1)| = \frac{3}{\sqrt{13}} = 0,832 \Rightarrow \left| T_2(j1) \right|_{dB} = 20 \log(0,832) = -1,6 \text{ dB}$$

$$|\alpha_2| = \frac{1}{|T_2|} = 1,6 \text{ dB} > 1 \quad \underline{\underline{\text{No cumple}}}$$

$$n=3$$

$$T_3(s) = \frac{(2,322)^2}{s + (2,322)^2} \cdot \frac{(2,542)^2}{s^2 + s \cdot 2,542 + 2,542^2}$$

0,691

$$|T_3(s=j1)| = \left| \frac{2,322}{j1 + 2,322} \cdot \frac{(2,542)^2}{s^2 + s \cdot 3,678 + 2,542^2} \right|$$

$$|T_3(s=j1)| = \left| \frac{2,322}{j1 + 2,322} \right| \left| \frac{(2,542)^2}{(j1)^2 + j1 \cdot 3,678 + 2,542^2} \right|$$

$$|T(j1)| = \frac{2,322}{\sqrt{2,322^2 + 1}} \cdot \frac{2,542^2}{\sqrt{(2,542^2 - 1^2)^2 + 3,678^2}}$$

$$|T(j1)| = 0,918 \cdot \frac{6,462}{6,584} = 0,901 \Rightarrow$$

$$|T(j1)|_{dB} = 20 \log |T(j1)| = 20 \log(0,901) = -0,905 dB$$

$$\Rightarrow |\alpha| = \frac{1}{|T|} \therefore \alpha = 0,905 dB < 1 \quad \underline{\text{cumple}}$$

$$n=4$$

$$T_4(s) = \frac{3,023^2}{s^2 + \frac{3,023}{0,5219} s + 3,023^2} \cdot \frac{3,389^2}{s^2 + \frac{3,389}{0,8055} s + 3,389^2}$$

$$T_4(s=j1) = \frac{3,023^2}{-1 + s \cdot 5,79 + 3,023^2} \cdot \frac{3,389^2}{-1 + s \cdot 4,207 + 3,389^2}$$

$$|T_4(j1)| = \frac{(3,023)^2}{\sqrt{(3,023^2 - 1)^2 + (5,79)^2}} \cdot \frac{3,389^2}{\sqrt{(3,389^2 - 1)^2 + (4,207)^2}}$$

$$|T_4(j1)| = \frac{3,023^2}{9,989} \cdot \frac{3,389^2}{11,297} = \frac{105}{112,845} = 0,930$$

$$|T_4(j1)|_{dB} = 20 \log |T(j1)| = 20 \log(0,930) = -0,626 dB$$

$$\alpha = 0,626 dB < 1 \quad \underline{\text{cumple}} \quad (\text{Menor atenuación})$$

el que cumple con el de mínimo orden es el de orden n=3

3 - retardo de grupo:

$$D(\omega) = -\frac{d\phi(\omega)}{d\omega}$$

$$n=2$$

$$T_2(s) = \frac{3}{s^2 + 3s + 3}$$

$$T_2(s=j\omega) = \frac{3}{-\omega^2 + j3\omega + 3} = \frac{3}{3 - \omega^2 + j3\omega}$$

$$\phi_{T_2} = \tan^{-1}\left(\frac{0}{3}\right) - \tan^{-1}\left(\frac{3\omega}{3 - \omega^2}\right)$$

$$\phi_{T_2} = -\tan^{-1}\left(\frac{3\omega}{3 - \omega^2}\right)$$

$$D_2(\omega) = +\frac{\partial}{\partial \omega} \left(\tan^{-1}\left(\frac{3\omega}{3 - \omega^2}\right) \right)$$

$$\frac{d}{d\omega} \tan^{-1}(\square) = \frac{1}{1 + \square^2} \cdot \square'$$

$$D_2(\omega) = \frac{1}{1 + \left(\frac{3\omega}{3 - \omega^2}\right)^2} \cdot \frac{3(3 - \omega^2) - 3\omega(-2\omega)}{(3 - \omega^2)^2}$$

$$\frac{(3 - \omega^2)^2}{(3 - \omega^2)^2 + 9\omega^2} \cdot \frac{9 - 3\omega^2 + 6\omega^2}{(3 - \omega^2)^2}$$

$$\frac{3\omega^2 + 9}{(3 - \omega^2)^2 + 9\omega^2} = \frac{3(\omega^2 + 3)}{9 - 6\omega^2 + \omega^4 + 9\omega^2}$$

$$D_2(\omega) = \frac{3(\omega^2 + 3)}{\omega^4 + 3\omega^2 + 9} = 0,4153$$

$$D_2(2,5) = \frac{3((2,5)^2 + 3)}{(2,5)^4 + 3(2,5)^2 + 9} = 0,4153$$

$$\rightarrow D(\omega) = -\sum_i \frac{\theta_i}{\theta_i^2 + (\omega - \omega_i)^2} - \sum_j \frac{\theta_j}{\theta_j^2 + (\omega - \omega_j)^2}$$

$$Q = \frac{\omega_0}{B\omega} \Rightarrow B\omega = \frac{\omega_0}{Q} = 3 \Rightarrow B\omega = 3 \wedge \omega_0 = \sqrt{3} \quad Q = \frac{\sqrt{3}}{3}$$

$$B\omega = 2\theta_j \Rightarrow \theta_j = \frac{B\omega}{2} \quad \therefore \theta_j = \frac{3}{2}$$

$$\omega_0^2 = \theta_j^2 + \omega_j^2 \Rightarrow \omega_j = \sqrt{\omega_0^2 - \theta_j^2} = \sqrt{3 - \left(\frac{3}{2}\right)^2} = \sqrt{\frac{12 - 9}{4}} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow D_2(\omega) = \frac{\theta_j}{\theta_j^2 + (\omega - \omega_j)^2} \Rightarrow \frac{3/2}{(3/2)^2 + (\omega - \frac{\sqrt{3}}{2})^2} + \frac{3/2}{(3/2)^2 + (\omega + \frac{\sqrt{3}}{2})^2}$$

$$D_2(2,5) = \frac{3/2}{(\frac{3}{2})^2 + (2,5 - \frac{\sqrt{3}}{2})^2} + \frac{3/2}{(\frac{3}{2})^2 + (2,5 + \frac{\sqrt{3}}{2})^2} = 0,4153 \checkmark$$

$$D_2(0) = 1 \Rightarrow \text{Error} = \frac{D_2(0) - D_2(2,5)}{D_2(0)} =$$

$$E\% = \frac{1 - 0,4153}{1} \cdot 100\% = 58,47\%$$

$$n=3$$

$$T_3(s) = \frac{(2,322)^2}{s + (2,322)^2} \cdot \frac{(2,542)^2}{s^2 + s \cdot \frac{2,542}{0,691} + 2,542^2}$$

$$\omega_{01} = 1,524$$

$$\omega_{02} = 2,542$$

$$Q_2 = 0,691$$

$$BW = 3,678$$

$$BW = 2\theta_j \Rightarrow \theta_j = BW/2 = 1,839$$

$$\omega_0^2 = \theta_j^2 + \omega_j^2 \Rightarrow \omega_j = \sqrt{\omega_0^2 - \theta_j^2} = 1,755$$

$$D_3(\omega) = \sum \frac{\theta_j}{\theta_j^2 + (\omega - \omega_j)^2}$$

$$D_3(\omega) = \frac{2,322}{2,322^2 + \omega^2} + \frac{1,839}{1,839^2 + (\omega - 1,755)^2} + \frac{1,839}{1,839^2 + (\omega + 1,755)^2}$$

$$D_3(2,5) = \frac{2,322}{2,322^2 + 2,5^2} + \frac{1,839}{1,839^2 + (2,5 - 1,755)^2} + \frac{1,839}{1,839^2 + (2,5 + 1,755)^2}$$

$$\frac{2,322}{11,64} + \frac{1,839}{3,937} + \frac{1,839}{21,487} = 0,1995 + 0,467 + 0,086 = 0,7525$$

$$E\% = \frac{1 - 0,7525}{1} \cdot 100\% = 24,75\%$$

$$n=4$$

$$T_4(s) = \frac{s^2 + \frac{3,023^2}{5,792} s + 3,023^2}{s^2 + 4,207 s + 3,389^2}$$

$$\omega_{01} = 3,023$$

$$Q_1 = 0,5219$$

$$BW = 5,792$$

$$BW = 2\theta_j$$

$$\theta_j = \frac{BW}{2} = 2,896$$

$$\omega_j = \sqrt{\omega_0^2 - \theta_j^2} = 0,867$$

$$\omega_{02} = 3,389$$

$$Q_2 = 0,8055$$

$$BW = 4,207$$

$$\theta_j = \frac{BW}{2} = 2,104$$

$$\omega_j = 2,657$$

$$D(\omega) = \frac{2,896}{2,896^2 + (\omega - 0,867)^2} + \frac{2,896}{2,896^2 + (\omega + 0,867)^2} + \frac{2,104}{2,104^2 + (\omega - 2,657)^2} + \frac{2,104}{2,104^2 + (\omega + 2,657)^2}$$

$$D(2,5) = \frac{2,896}{2,896^2 + (2,5 - 0,867)^2} + \frac{2,896}{2,896^2 + (2,5 + 0,867)^2} + \frac{2,104}{2,104^2 + (2,5 - 2,657)^2} + \frac{2,104}{2,104^2 + (2,5 + 2,657)^2}$$

$$\frac{2,896}{11,054} + \frac{2,896}{19,724} + \frac{2,104}{4,451} + \frac{2,104}{31,021} = 0,262 + 0,147 + 0,472 + 0,068$$

$$D_3(2,5) = 0,949$$

$$\varepsilon\% = \frac{D(0) - D_3(2,5)}{D(0)} \cdot 100\% = \frac{1 - 0,949}{1} \cdot 100\% \approx 5,1\%$$

4- Sintetizar el circuito

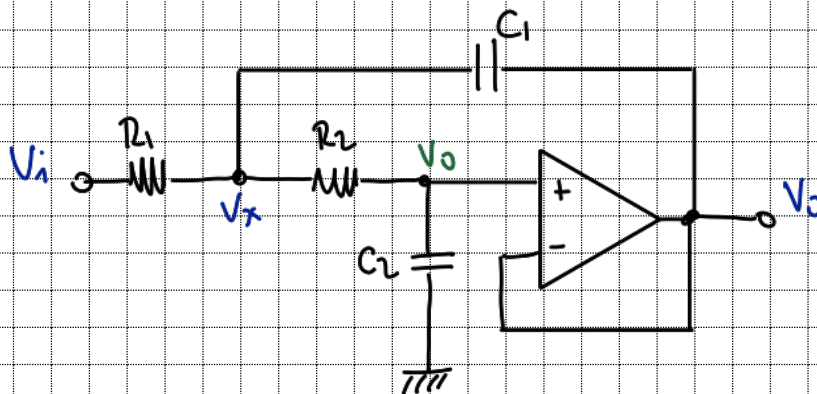
$$T(s) = \frac{2,322}{s + 2,322} \cdot \frac{6,462}{s^2 + s \cdot 3,678 + 6,462} \quad Q = 0,691$$

① ②

① $\frac{2,322}{s + 2,322} \quad \frac{1}{RC} = 2,322 \Rightarrow R = \frac{1}{C \cdot 2,322} \quad C = 1 \Rightarrow R = 0,431$

② $\frac{6,462}{s^2 + s \cdot 3,678 + 6,462}$

Análisis del circuito



$$\begin{cases} V_x(G_1 + G_2 + sC_1) = V_i G_1 + V_o(sC_1 + G_2) \\ V_o(G_2 + sC_2) = V_x G_2 \rightarrow V_x = R_2(G_2 + sC_2) V_o \\ V_x = V_o(1 + sR_2C_2) \end{cases}$$

$$V_o(1 + sR_2C_2)(G_1 + G_2 + sC_1) - V_o(sC_1 + G_2) = V_i G_1$$

$$V_o(G_1 + G_2 + sC_1 + sR_2C_2(G_1 + G_2) + s^2R_2C_1C_2 - sC_1 - G_2)R_1 = V_i$$

$$\frac{V_o}{V_i} = T(s) = \frac{1}{s^2R_1R_2C_1C_2 + sR_1R_2C_2(G_1 + G_2) + R_1G_1}$$

$$T(s) = \frac{1}{s^2R_1R_2C_1C_2 + sR_1R_2C_2 \frac{R_1 + R_2}{R_1R_2} + \frac{R_1}{R_1}}$$

$$T(s) = \frac{1}{s^2R_1R_2C_1C_2 + sC_2(R_1 + R_2) + 1}$$

$$T(s) = \frac{1}{R_1R_2C_1C_2} \cdot \frac{1}{s^2 + s \frac{(R_1 + R_2)}{R_1R_2C_1} + \frac{1}{R_1R_2C_1C_2}}$$

$$T(s) = \frac{1}{R_1 R_2 C_1 C_2} \cdot \frac{1}{s^2 + s \frac{(R_1 + R_2)}{R_1 R_2 C_1} + \frac{1}{R_1 R_2 C_1 C_2}}$$

$$\omega_0^2 = \frac{1}{R_1 R_2 C_1 C_2} \Rightarrow \omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$

$$\frac{\omega_0}{Q} = \frac{(R_1 + R_2)}{R_1 R_2 C_1} \Rightarrow \frac{R_1 R_2 C_1}{(R_1 + R_2)} \omega_0 = Q$$

$$Q = \frac{R_1 R_2 C_1}{\sqrt{R_1 R_2 C_1 C_2}} \cdot \frac{1}{(R_1 + R_2)} \Rightarrow Q = \sqrt{\frac{R_1 R_2 C_1}{C_2}} \cdot \frac{1}{R_1 + R_2} \cdot \frac{C_1}{C_1} \Rightarrow Q = \frac{\sqrt{R_1 R_2 C_1 C_1}}{(R_1 + R_2) C_1}$$

• Elijo $T(s)$ de orden $n=3$

$$T_3(s) = \frac{2,322}{s + 2,322} \cdot \frac{(2,542)^2}{s^2 + s \cdot \frac{2,542}{0,691} + 2,542^2}$$

$$\omega_0 = 2,322$$

$$\frac{1}{RC} = 2,322 \quad R=1$$

$$C = \frac{1}{2,322}$$

$$\bullet \quad \frac{1}{R_1 R_2 C_1 C_2} = 2,542^2 = \omega_0^2$$

$$\bullet \quad Q = \frac{1}{\omega_0 (R_1 + R_2) C_1}$$

$$R_1 R_2 C_1 C_2 = \frac{1}{2,542^2} \quad \wedge \quad R_1 R_2 C_1 C_2 = 2,542^2, \quad R_1 = R_2 = 1$$

$$\Rightarrow Q = \frac{1}{(2,542)^2 C_2} \Rightarrow C_2 = \frac{1}{(2,542)(0,691)^2} = 0,2846 = C_2$$

$$C_1 = \frac{1}{R_1 R_2 C_2 (2,542)^2} = \frac{1}{(2,542)^2 \cdot 0,2846} = 0,5436 = C_1$$

$T_N(s)$: Normalizado

$$T_3(s) = \frac{2,322}{s + 2,322} \cdot \frac{(2,542)^2}{s^2 + s \cdot \frac{2,542}{0,691} + 2,542^2}$$

$$s = 2,542s$$

$$\Rightarrow T_{3N}(s) = \frac{2,542^2}{2,542^2 s^2 + 2,542 s \cdot 2,542 + 2,542^2}$$

$$T_{3N}(s) = \frac{1}{s^2 + s \cdot \frac{1}{0,691} + 1}$$

$$\Rightarrow R_1 R_2 C_1 C_2 = 1 \quad \wedge \quad \sqrt{R_1 R_2 C_1 C_2} = 1, \quad R_1 = R_2 = 1 \quad C_1 = \frac{1}{C_2}$$

$$Q = \frac{1}{2C_2} \Rightarrow C_2 = \frac{1}{2Q} = 0,7236$$

$$C_1 = 1,382$$

$$s = 2,322s$$

$$\Rightarrow \frac{2,322}{s \cdot 2,322 + 2,322} \Rightarrow \frac{1}{s+1} \Rightarrow 1 = \frac{1}{RC} \quad \wedge \quad R=1$$

$$\Rightarrow C=1$$

$$\Rightarrow |T_{3N}(s=j\omega)| = \left| \frac{1}{j\omega+1} \cdot \frac{1}{1+j\omega \cdot \frac{1}{0,691} + 1} \right| \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\frac{1}{0,691}}$$

$$|T_{3N}(j\omega)| = \frac{1}{\sqrt{2}} \cdot 0,691 = 0,4886$$

$$\left| T_{3N}(j\omega) \right|_{dB} = 20 \log |T_{3N}(j\omega)| = 20 \log(0,4886) = -6,2dB$$